Resilience-Aware, Human-Centric AI-RAN Orchestration for Service Continuity Across Terrestrial and Non-Terrestrial Networks

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Abstract—We describe a methods-ready research artifact for privacy-preserving, phone-first field measurement and twin-informed AI-RAN orchestration under heterogeneous terrestrial and non-terrestrial connectivity. The artifact couples canonical JSON Schema contracts, an integrated Gate 2 pipeline, preregistered evaluation design, and a 54-cell controlled pilot matrix (3 zones $\times$ 2 network conditions $\times$ 3 days $\times$ 3 workload profiles). We explicitly disclaim carrier-grade AI-RAN, deployable 6G infrastructure, citywide generalization, and unauthorized RF transmission. Primary outcome total_service_outage_time_s (secondary time_to_recovery_s; Amendment 001 in progress) is defined from physical probes; empirical pilot results are RESULTS_PENDING_AUTHENTIC_GATE3_DATA with no outcome numbers reported here.

Index Terms—6G-aligned, IMT-2030, AI-RAN, O-RAN, digital twin, non-terrestrial networks, reproducibility, privacy-preserving measurement

I. Introduction

Sparse, degraded, and disrupted connectivity remains a practical barrier to service continuity for learning, creation, and sensing workloads in under-connected communities. IMT-2030-aligned research explores AI-native air interfaces, open RAN disaggregation, and multi-access resilience including non-terrestrial network (NTN) fallback [1]–[3]. Operational claims about carrier-grade AI-RAN or deployable 6G infrastructure are outside the scope of this artifact.

This paper documents the methods-ready umbrella repository gunnchos-7gc-ai-ran-field-kit, which locks a doctoral thesis hypothesis and three research questions (RQ1–RQ3) in Gate 1, provides executable contracts and orchestration in Gate 2, defines controlled evidence collection in Gate 3, and preregisters scientific evaluation in Gate 4. We separate engineering completion from release evidence: Gate 2 system pass does not imply immutable archive, DOI deposit, or non-author reproduction.

A. Central hypothesis

A privacy-preserving orchestration architecture that combines physical device measurements, workload and service intent, digital-twin context, and terrestrial, local-edge, and NTN state can improve service continuity and worst-user quality of experience under sparse connectivity when compared with static, network-only, and service-priority baselines, while maintaining explicit fairness, energy, privacy, and reliability constraints.

B. Research questions

RQ1 (Trustworthy context): How can privacy-preserving Edge-IO measurements be transformed into reproducible digital-twin state without direct identifiers? RQ2 (Human-centric orchestration): To what extent does twin-informed AI-RAN orchestration improve continuity and worst-user QoE versus defined baselines? RQ3 (Multi-access resilience): Under which outage, latency, and privacy conditions do terrestrial, offline, and NTN fallback policies help or fail?

C. Contributions (methods only)

- Canonical cross-repository JSON Schema contracts and provenance for Edge-IO $\rightarrow$ twin $\rightarrow$ AI-RAN $\rightarrow$ NTN resilience paths.
- A preregistered statistical analysis plan with primary outcome total_service_outage_time_s (Amendment 001, in progress) and secondary time_to_recovery_s; baseline registries frozen before authentic pilot inspection.
- A 54-cell pilot design matrix with explicit consent, privacy scan, and session-mode rules separating calibration, rehearsal, and eligible pilot sessions.
- Honest claim boundaries: synthetic Gate 4 infrastructure results are labeled separately from pending Gate 3 field evidence.

Outcome tables and effect sizes are withheld until authentic Gate 3 pilot data exist (RESULTS_PENDING_AUTHENTIC_GATE3_DATA).

II. Related Work

A. IMT-2030 and 6G vision

ITU-R Recommendation M.2160 defines the framework and overall objectives for IMT-2030 development [1]. Survey literature outlines AI-native air interfaces, extreme connectivity, and integrated sensing and communication as research directions rather than deployed capabilities [2], [3]. We use 6G-aligned and IMT-2030-aligned wording only.

B. Open RAN and AI-RAN experimentation

The O-RAN Alliance architecture disaggregates the RAN into open interfaces suitable for policy and ML experimentation [4]. This artifact implements AI-assisted O-RAN-style control experiments in simulation; it does not claim near-real-time RIC deployment on commercial networks.

C. Edge computing and measurement

ETSI MEC defines a framework for edge-hosted applications and orchestration [5]. Companion work in edge-io-measurement-node specifies opt-in, schema-validated telemetry without payload capture; we reference that schema rather than redefining it here.

D. Non-terrestrial networks

3GPP TR 38.821 documents NR solutions for NTN scenarios including satellite and aerial platforms [6]. Our resilience baselines (terrestrial_only, always_ntn_on_terrestrial_failure, etc.) are evaluated in source-validated simulation paths; live NTN control is not claimed.

E. Reproducibility and preregistration

Preregistered outcomes, holdout registries, and negative-result preservation follow open-science reproducibility practices documented in evaluation/EVALUATION_PREREGISTRATION.md (formal preregistration citation: CITATION_NEEDED). This repository locks PRIMARY_OUTCOME_LOCK.json and EVALUATION_PREREGISTRATION.md before complete pilot result inspection.

III. System Architecture and Contracts

A. Umbrella orchestration role

gunnchos-7gc-ai-ran-field-kit is the canonical contract authority linking:

- edge-io-measurement-node — privacy-preserving device telemetry (RQ1),
- 7gc-digital-twin — twin-state construction (RQ1–RQ2),
- spectrumx-ai-ran-gary — AI-RAN policy baselines and ablations (RQ2–RQ3),
- ntn-resilience-sim — multi-access resilience simulation (RQ3).

Optional PHY supporting studies (e.g., ReadyGary beam selection) may inform assumptions but are not required for thesis completion.

B. Gate model

TABLE I
Research gates and evidentiary meaning

Gate	Meaning
Gate 1	Locked thesis, hypothesis, RQ1–RQ3
Gate 2	Executable schemas + integrated pipeline
Gate 3	Authentic controlled-device pilot evidence
Gate 4	Preregistered evaluation on eligible evidence
Gate 5/6	Release archive, DOI, non-author reproduction

C. Canonical schemas

Contracts under contracts/ include measurement session context, Edge-IO batches, twin bundles, AI-RAN decision bundles, and Gate status reports. All integrated runs emit SHA-256 provenance via scripts/verify_provenance.py. Invalid fixtures under fixtures/invalid/ document rejection reasons for contract tests.

D. Integrated pipeline

The default reproduction path executes:

```
make test && make integrated-pipeline \
  EDGE_INPUT=fixtures/valid/edge_measurement_batch.valid.json
```

Outputs land under results/integrated/ with evidence labels synthetic or controlled_device_measurement. Synthetic outputs support Gate 4 infrastructure readiness only; they must not be presented as field pilot results.

IV. System Model and Notation

TABLE II
Core notation (methods manuscript)

Symbol	Meaning
$\mathcal{U}$	set of participant devices (clustered by day/zone)
$\mathcal{S}$	service/workload profiles (learn, create, sense)
$\mathcal{M}$	access modes: terrestrial, local edge, NTN, offline
$\mathbf{x}_t$	Edge-IO probe at timestamp t (latency, flags, intent)
$\mathcal{T}_t$	digital-twin state bundle at t (identifier-free)
π	AI-RAN policy over RAN-style actions
ρ	multi-access resilience policy over $\mathcal{M}$
$\mathcal{C}$	pilot matrix cell (z, d, c, w): zone, day, condition, workload

We formalize the artifact as a closed-loop measurement–twin–policy pipeline over discrete probe intervals. Table II summarizes symbols used throughout the manuscript. All empirical quantities derived from device probes are distinguished from model-estimated fields such as expected_recovery_time_s in resilience decision bundles.

A. Session and matrix structure

Each eligible pilot session s is indexed by matrix cell $\mathcal{C}_s \in \mathcal{Z} \times \mathcal{D} \times \mathcal{N} \times \mathcal{S}$ with $|\mathcal{Z}| = 3$ zones, $|\mathcal{D}| = 3$ days, $|\mathcal{N}| = 2$ network conditions, and $|\mathcal{S}| = 3$ workload profiles (54 cells total). Sessions with session_mode=PILOT after consent, privacy scan, and assignment-hash verification contribute to Gate 3 evidence; calibration and rehearsal sessions are excluded from the matrix count.

B. Probe timeline and availability

For session s, ordered probes $\{\mathbf{x}_t\}_{t=1}^{T_s}$ are emitted by edge-io-measurement-node at workload-defined cadence (minimum 50 samples over 300 s per workload protocol). Availability at probe t is a Boolean function $a(\mathbf{x}_t)$:

$$a(\mathbf{x}_t) = \begin{cases} \text{service\_available}, & \text{if field present} \\ \neg\text{probe\_timeout} \wedge (\text{latency\_ms} \neq \emptyset), & \text{otherwise.} \end{cases} \quad (1)$$

Outage intervals begin at the first unavailable probe after availability and end at the first subsequent available probe, or continue until session end (right censoring).

C. Primary and secondary outcomes

The amended primary outcome is total_service_outage_time_s: the sum of outage-interval durations within session s (lower is better). Amendment 001 documents the change from the prior label recovery_time_s, which conflated total outage duration with a single recovery-time estimand; the amendment is internally valid and awaiting final author approval before physical collection. The secondary outcome is time_to_recovery_s at the event level: duration from outage onset to first available probe for completed recoveries only. Outages still active at the final probe are right-censored and excluded from completed-recovery summaries (no imputation as observed recoveries). Physical computation follows scripts/compute_recovery_time.py on frozen edge_measurement_batch schema fields.

D. Decision pipeline

The integrated Gate 2 path maps $\mathbf{x}_t \rightarrow \mathcal{T}_t \rightarrow \pi(\mathcal{T}_t) \rightarrow \rho(\mathcal{T}_t, \pi)$, producing schema-validated airan_decision_bundle and resilience_decision_bundle artifacts with SHA-256 provenance. AI-RAN policies π select RAN-style actions (resource blocks, scheduling, compute placement); resilience policies ρ select selected_mode $\in \mathcal{M}$ subject to contract-recorded constraints and rejected-alternative audit trails.

V. Orchestration Objectives, Constraints, and Multi-Access Logic

A. Service-continuity objective

Twin-informed orchestration targets service continuity: sustaining workload-appropriate connectivity and local-edge responsiveness under degraded conditions. The AI-RAN layer optimizes predicted continuity-related metrics (service_continuity_score, scheduling and placement actions) subject to explicit constraint violations recorded in constraint_violations. The resilience layer selects among terrestrial, local-edge, NTN, and offline continuation modes to maximize continuity_score while respecting energy, privacy, and feasibility gates encoded in rejected_alternatives. Model outputs such as expected_recovery_time_s are not substituted for physical total_service_outage_time_s; they support policy comparison and simulation only until Gate 3 evidence is frozen.

B. Fairness, energy, reliability, and privacy constraints

Constraints are enforced at decision time and audited in bundle schemas:

- Fairness: AI-RAN bundles expose fairness_index and ablation fairness_constraint toggles worst-user protection versus uniform allocation; violations append to constraint_violations.
- Energy: Resilience bundles record estimated_energy_cost_j; ablation energy_constraint removes the energy ceiling from candidate ranking.
- Reliability: Predicted reliability derives from loss models (reliability_from_loss: $1 -$ predicted_packet_loss_pct/100) and informs mode rejection when below policy thresholds.
- Privacy: estimated_privacy_cost $\in [0, 1]$ ranks alternatives; telemetry remains identifier-free per Edge-IO schema; ablation remove_privacy_constraint_simulated_decision_constraint_only is a simulated toggle—live privacy controls stay enabled during collection.

Ablation registry entries (physical_telemetry, digital_twin_context, continuity_objective, local_edge_option, ntn_option, workload_intent) isolate component contributions without inventing outcome numbers here.

C. Terrestrial, edge, and NTN decision logic

Resilience policy ρ evaluates candidates in registry order for baseline service_aware_multi_access (proposed path):

1) Terrestrial: default when predicted latency and capacity satisfy workload minima and terrestrial link state is nominal.
2) Local edge (local_edge_wifi, degraded_local): considered when terrestrial probes indicate elevated latency or loss but a local-edge endpoint responds within local_edge_response_ms budgets recorded in Edge-IO batches.
3) NTN fallback (ntn_fallback): eligible when terrestrial and local-edge paths fail triggers (timeout, capacity floor, or policy-specific fallback_trigger), and NTN assumptions in the bundle pass energy and privacy gates.
4) Offline continuation (offline_continuation, delayed_synchronization): selected when no radio path meets continuity minima; retained versus suspended service features are enumerated for audit.

Baselines terrestrial_only, terrestrial_then_offline, always_ntn_on_terrestrial_failure, and priority_class_fallback instantiate strict subsets of this candidate set for controlled comparison. oracle_hindsight_analysis_only is analysis-only and never deployable.

Algorithm 1 Service-aware multi-access resilience selection (template)

Require: twin state $\mathcal{T}$, AI-RAN decision d, policy registry entry P

Ensure: resilience bundle with selected_mode, audit fields

$\mathcal{C} \leftarrow$ BuildCandidates$(\mathcal{T}, d, P)$ {terrestrial, edge, NTN, offline}
for each candidate $m \in \mathcal{C}$ **do**
  $(\hat{L}_m, \hat{C}_m, E_m, R_m, \Pi_m) \leftarrow$ EstimateMetrics$(m, \mathcal{T})$
  if ViolatesFairness(m) or ViolatesEnergy(m, E_m) **then**
    Reject$(m$, reason); **continue**
  end if
  if ViolatesReliability(m, R_m) or ViolatesPrivacy(m, Π_m) **then**
    Reject$(m$, reason); **continue**
  end if
  score$(m) \leftarrow$ ContinuityScore$(m, \hat{L}_m, \hat{C}_m$, workload intent)
end for
$m^\star \leftarrow \arg\max_{m \in \mathcal{C} \setminus \text{rejected}}$ score(m)
return bundle(selected_mode=$m^\star$, rejected_alternatives, hashes)

D. Policy pseudocode

Algorithm 1 summarizes the service-aware multi-access template implemented in ntn-resilience-sim (deterministic given inputs and policy name).

AI-RAN twin-informed policy π_{twin} extends network-only scoring with twin-derived features (zone context, probe cadence, workload class) while sharing the same constraint audit structure; static and network-only baselines skip twin features; service-priority elevates continuity class weights; optimization-based runs constrained search.

E. Computational complexity

Let $|\mathcal{C}| \leq 4$ access modes and $|\mathcal{A}|$ denote AI-RAN discrete action facets (resource blocks, scheduling tiers) bounded by fixture registry sizes. Per integrated run, AI-RAN policy evaluation is $O(|\mathcal{A}|)$ under tabulated baseline implementations; resilience selection is $O(|\mathcal{C}|)$ with constant-factor metric estimation. Twin construction is $O(T_s)$ in probe count for session s. Gate 4 batch evaluation over B baselines and A ablations is $O(B \cdot A \cdot T_s)$ for metric recomputation-dominated workloads—tractable for the 54-cell pilot matrix on commodity hardware; decision runtime_s and peak_memory_mb are logged per bundle for empirical complexity verification (no numeric results reported here).

VI. Measurement and Pilot Protocol

A. Evidence tiers

We distinguish smoke CI tests, synthetic reproducible benchmarks, controlled device measurements, and operational RAN claims (the last is not claimed).

B. Physical measurement methodology

Gate 3 evidence derives from phone-first Edge-IO probes validated against contracts/edge_measurement_batch.v1.schema.json. Each sample records ISO-8601 timestamp, optional latency_ms, quality_flags (e.g., probe_timeout), optional explicit service_available, workload and service profile identifiers, and local_edge_response_ms when probed. Probes are sorted temporally; consecutive timestamps define observation intervals. Unavailable classification follows the precedence in Section IV. The metric engine (scripts/compute_recovery_time.py) constructs outage events, computes total_service_outage_time_s, emits per-event time_to_recovery_s records, flags right censoring, and reports probe cadence statistics (median interval, missing-probe gaps >2.5× median). Interval-censoring uncertainty is bounded by ± half the median probe interval—true transition times may fall between probes. Model fields such as expected_recovery_time_s in resilience bundles are never merged into physical outcomes.

C. Session modes

Only sessions with session_mode=PILOT count toward the 54-cell matrix after consent, privacy scan, and assignment-hash verification. Calibration and rehearsal sessions support infrastructure validation and are explicitly excluded from pilot counts.

D. Pilot matrix design

The matrix spans:

- 3 zones (zone_a, zone_b, zone_c) with public labels pending human approval,
- 2 network conditions (wifi_normal, wifi_degraded) applied via scripted condition scripts,
- 3 collection days (day_01–day_03),
- 3 workload profiles (learn, create, sense), each 300 s.

Authoritative cell definitions live in pilot/54_CELL_ASSIGNMENT_MATRIX.csv. At manuscript freeze: 0/54 eligible cells collected; HUMAN_ACTION_REQUIRED.

E. Workload protocols

Each workload specifies periodic request cadence, timeout handling, and success criteria (duration ≥300 s, samples ≥50, explicit nulls for unavailable metrics). Details are in pilot/WORKLOAD_PROTOCOLS.md.

F. Network condition protocol

Condition application and restoration use pilot/scripts/apply_condition.sh, verify_condition.sh, and restore_condition.sh with parameters in pilot/configs/network_conditions.yaml. Deviations are logged in datasets/controlled/protocol-deviations/PROTOCOL_DEVIATION_LOG.md.

G. Privacy and ethics

Collection follows opt-in consent, no payload inspection, no direct identifiers, aggregated public reporting, and receive-only RF boundaries where SDR observation is discussed (see docs/PRIVACY_AND_ETHICS.md; formal IRB citation: CITATION_NEEDED). Raw device exports remain in gitignored datasets/controlled/raw-private/; sanitized derivatives require human approval before publication. Ethics documentation does not imply regulatory certification or law-enforcement monitoring capabilities.

VII. Evaluation Design and Statistical Plan

A. Primary claim and outcomes

The preregistered primary claim states that twin-informed service-aware orchestration reduces service outage duration relative to static, network-only, and service-priority policies under defined degraded connectivity, subject to energy, fairness, privacy, and reliability constraints. Primary outcome: total_service_outage_time_s—total seconds the service is classified unavailable within an eligible session (lower is better), computed from physical Edge-IO probes via scripts/compute_recovery_time.py. Secondary outcome: time_to_recovery_s—per-outage recovery duration for completed recoveries only. Amendment 001 supersedes the prior primary label recovery_time_s, which summed unavailable intervals but was named like a single recovery-time estimand; the original lock remains byte-preserved under evaluation/amendments/amendment_001/ pending final author approval before collection. Practical significance for the primary metric uses a 5 s minimal interesting difference (frozen with preregistration thresholds).

B. Baseline rationale

Baselines are registered before authentic result inspection to prevent post-hoc policy selection:

- static_uniform — equal allocation ignoring twin and service intent (lower bound on adaptivity).
- network_only — reacts to network KPIs without twin context (isolates RQ1 contribution).
- service_priority — elevates continuity class without twin features (isolates intent-only path).
- optimization_based — constrained search baseline for RQ2 comparison.
- twin_informed — proposed AI-RAN policy using identifier-free twin state.

Resilience baselines span terrestrial_only through service_aware_multi_access, with oracle_hindsight_analysis_only reserved for upper-bound analysis only. Registries: evaluation/BASELINE_REGISTRY.yaml, evaluation/ABLATION_REGISTRY.yaml, evaluation/HOLDOUT_REGISTRY.yaml.

C. Holdout and ablation rationale

Leave-one-out holdouts (leave_one_day_out, leave_one_zone_out, leave_one_condition_out, leave_one_workload_out) test generalization across matrix axes without treating 54 sessions as independent participants. stress_scenario_holdout excludes configured stress scenarios from primary fitting (evaluation/configs/held_out_scenarios.yaml). Ablations remove one component at a time (telemetry, twin, continuity objective, fairness, energy, local edge, NTN, workload intent) to attribute failures and neutral results. Split leakage checks and duplicate detection are mandatory before any primary-outcome inference (evaluation/LEAKAGE_AND_DUPLICATE_REPORT.md).

D. Statistics, clustering, and censoring

Analysis follows evaluation/STATISTICAL_ANALYSIS_PLAN.md:

- Grouped summaries by day and zone with cluster-aware treatment of repeated measures.
- Bootstrap and repeated-seed uncertainty (95% CI, 1000 resamples, seed 0 per evaluation/configs/metrics.yaml).
- Effect sizes and practical significance against PRACTICAL_SIGNIFICANCE_THRESHOLDS.yaml.
- Missing-data and protocol-deviation audits (evaluation/MISSING_DATA_ANALYSIS.md).

Censoring treatment: Outages ongoing at session end are right-censored. Censored events contribute to total_service_outage_time_s (duration to last probe) but do not supply observed time_to_recovery_s values. Summary statistics for the secondary metric exclude censored events; sensitivity analyses may report censoring rate and interval-censoring bounds (± half median probe interval) without imputing completed recoveries. Negative and neutral results are preserved per evaluation/NEGATIVE_AND_NEUTRAL_RESULTS.md.

E. Synthetic Gate 4 infrastructure

Gate 4 automation demonstrates evaluation plumbing on synthetic or fixture-backed inputs. Those runs are labeled GATE4_EVALUATION_READY infrastructure, not authentic pilot completion. No numeric results from Gate 4 are imported into this methods manuscript.

VIII. Results

RESULTS_PENDING_AUTHENTIC_GATE3_DATA

Authentic controlled-device pilot sessions required for primary-outcome analysis have not been collected at manuscript freeze. The eligible pilot matrix status is 0/54 cells. Calibration evidence (e.g., one valid Pixel 6a calibration session) supports infrastructure validation only and must not be used to claim pilot completion or causal superiority.

When authentic Gate 3 data exist, this section will report:

- pilot coverage against pilot/54_CELL_ASSIGNMENT_MATRIX.csv,
- primary outcome total_service_outage_time_s and secondary time_to_recovery_s (with censoring rates) by baseline and holdout split,
- uncertainty intervals, effect sizes, and practical significance against locked thresholds,
- negative and neutral results per evaluation/NEGATIVE_AND_NEUTRAL_RESULTS.md,
- failure boundaries per evaluation/FAILURE_BOUNDARIES.md.

No tables, figures, p-values, or effect sizes are reported here to avoid inventing empirical numbers.

IX. Limitations, Ethics, and Threats to Validity

A. Scope limits

This artifact is a research prototype, not commercial 6G, carrier-grade AI-RAN, or citywide deployment infrastructure. Gary/NWI anchor context supports digital-equity motivation but does not justify global generalization.

B. Evidence limits

Synthetic benchmarks and Gate 4 infrastructure runs demonstrate reproducible engineering and evaluation plumbing only. Until Gate 3 authentic pilot data exist, all outcome claims remain RESULTS_PENDING_AUTHENTIC_GATE3_DATA.

C. Design threats

Repeated measures within days and zones induce clustering; the statistical plan addresses this but cannot remove locale-specific confounds. Network condition scripts approximate degradation; they do not emulate full RF propagation diversity. NTN resilience paths are simulation-backed unless and until validated against controlled NTN-capable environments. Interval censoring from finite probe cadence bounds temporal precision of outage onsets and recoveries.

D. Privacy, consent, and human subjects

Structured field campaigns require IRB or equivalent approval before identifiable participant collection (CITATION_NEEDED). Consent versions, opt-in boundaries, and deletion/retention policies are documented in docs/PRIVACY_AND_ETHICS.md. Public release depends on sanitized derivatives and consent-summary approval. Missing cells, protocol deviations, and consent withdrawals must be reported rather than imputed without documented sensitivity analysis. Simulated privacy-constraint ablations never disable live telemetry protections.

E. Reproducibility limits

Clean-checkout author reproduction, non-author reproduction, immutable release tarball, and DOI deposit remain Gate 5/6 requirements and are pending at manuscript freeze. PDF builds use pinned tooling (scripts/build_paper.sh) with SOURCE_DATE_EPOCH=0 for reproducibility where supported; residual PDF byte differences may arise from hyperref object IDs or tool-specific metadata (documented in PAPER_BUILD_REPORT.md).

X. Conclusion

We presented a methods-ready, honesty-bounded artifact for resilience-aware, human-centric AI-RAN orchestration research spanning privacy-preserving measurement, twin-informed policy evaluation, and NTN fallback simulation. Gate 1 thesis elements, Gate 2 contracts, Gate 3 pilot protocol, and Gate 4 preregistration are in place; empirical pilot outcomes remain RESULTS_PENDING_AUTHENTIC_GATE3_DATA. The amended primary outcome total_service_outage_time_s with secondary time_to_recovery_s aligns physical probe semantics with preregistered claims (Amendment 001, approval pending).

Future work completes the 54-cell matrix under human-approved zones and dates, executes the locked statistical analysis plan on eligible evidence only, and pursues release/archive steps with independent reproduction before outcome claims appear in a results manuscript.

References

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Appendix A
Reproducibility Appendix

A. Environment

See repository ENVIRONMENT.md for Python version, dependency pins, and sibling repository layout under

REPOS_ROOT. Integrated runs require sibling repos edge-io-measurement-node, 7gc-digital-twin, spectrumx-ai-ran-gary, and ntn-resilience-sim at commits recorded in PROVENANCE_LOCK.json.

B. Author reproduction (pending)

AUTHOR_REPRODUCTION_REPORT.md records a clean-checkout run outcome; status at freeze: PENDING.

C. Non-author reproduction (pending)

NON_AUTHOR_REPRODUCTION_REPORT.md is an authentic empty pending template until an independent replicator files results.

D. Core commands

```
pip install -r requirements.txt
make test
make contract-test
make integrated-pipeline \
  EDGE_INPUT=fixtures/valid/edge_measurement_batch.valid.json
make gate1-validate
python3 scripts/compute_recovery_time.py \
  --input fixtures/valid/edge_measurement_batch.valid.json
```

E. Outcome metric reproduction

Physical outcomes are recomputed from Edge-IO batches only. The CLI above emits total_service_outage_time_s, event-level time_to_recovery_s, censoring flags, and probe cadence diagnostics without referencing model expected_recovery_time_s fields.

F. Pilot collection (human-gated)

Pilot counting requires pilotctl.py workflows, approved zone/date tokens, and storage under datasets/controlled/raw-private/ (gitignored). Public artifacts document protocol only until sanitized publication approval.

G. Paper build reproducibility

PDF build uses pinned Tectonic (scripts/build_paper.sh, version pinned in-script) or a digest-pinned TeX Live Docker image; local pdflatex/bibtex remains an optional fallback. Builds export SOURCE_DATE_EPOCH=0 (and TZ=UTC) before compilation to stabilize timestamps where tools honor them. Unavoidable byte-level PDF differences may still occur from hyperref anchor generation, filesystem XMP metadata, or Tectonic cache state; PAPER_BUILD_REPORT.md records page count (required 6–8 inclusive), SHA-256, and build method each run. See paper/Makefile and paper/README.md.